LiDAR Signal Fusion Method Based on AI-Driven Saturation Detection and Nonlinear Calibration

Long Xu, Xin Lin, Yong Yang, Linmei Liu, Wei Wang, Xuewu Cheng, Kaijie Ji, Jiqin Wang, and Faquan Li

***Abstract*—Ground-based dual-channel Rayleigh LiDAR provides continuous observations of the middle and upper atmosphere, but reliable full-altitude profiling remains difficult. Strong near-range echoes may drive the low-altitude channel into nonlinear compression or saturation, whereas weak high-altitude echoes are often affected by severe signal-to-noise-ratio degradation. Mismatch between the low- and high-altitude channels can also introduce discontinuities in the overlap region, which may propagate into subsequent density and temperature retrievals. Existing studies have mainly addressed detector-level dead-time correction or local channel stitching, but the complete processing chain from saturation detection and nonlinear correction to overlap-region consistency and retrieval-oriented fusion has not been fully considered. To address this limitation, we propose an AI-driven LiDAR signal fusion framework that integrates saturation detection, nonlinear calibration, and retrieval-consistent channel stitching. An anomaly transformer (AT) is first used to identify the onset of nonlinear response and determine the valid fusion interval. A nonparalyzable detector model combined with extreme gradient boosting (XGBoost) residual learning is then applied to correct saturation-affected segments. Finally, an interchannel nonlinear mapping model and a three-stage weighted stitching strategy are used to generate continuous fused profiles before physical inversion. Experiments on both isomorphic and heterogeneous photomultiplier-tube (PMT) systems show that the proposed framework improves signal continuity and retrieval consistency. For the isomorphic configuration, it improves temperature-profile consistency within the overlap region, reducing the maximum temperature uncertainty from 4.8 to 3.7 K and the root-mean-square error (RMSE) from 2.45 to 1.62 K. For the heterogeneous configuration, the retrieved temperature profile agrees well with the reference model over 30–80 km, with deviations at most altitudes confined within $\pm$3 K and a maximum deviation of approximately 3.8 K. These results demonstrate the robustness of the proposed framework for retrieval-oriented dual-channel Rayleigh LiDAR signal fusion under both isomorphic and heterogeneous PMT conditions.**

***Index Terms*—Extreme gradient boosting (XGBoost) residual learning, nonlinear correction, Rayleigh LiDAR, saturation detection, signal stitching, temperature retrieval.**

I. Introduction

THE middle and upper atmosphere is a key region for energy exchange, wave dissipation, and thermodynamic coupling. Its thermal structure provides important information for understanding gravity-wave propagation, atmospheric tides, and space-weather-related variability. Owing to its high vertical resolution and continuous nighttime observation capability, ground-based Rayleigh LiDAR has become an effective tool for retrieving atmospheric density and temperature in this altitude range. However, continuous full-altitude profiling remains challenging. In dual-channel LiDAR systems, strong near-range returns may drive the low-altitude channel into nonlinear compression or saturation, whereas weak high-altitude echoes are often affected by low signal-to-noise ratio. The resulting mismatch in the overlap region can produce amplitude jumps or slope discontinuities after channel stitching, and these fusion-induced distortions may further propagate into density and temperature retrievals. Robust saturation-aware nonlinear correction and retrieval-consistent signal fusion are, therefore, essential for improving the reliability of middle- and upper-atmospheric LiDAR observations.

A representative example of direct dual-channel stitching is shown in Fig. 1. Although both channels retain the expected altitude-dependent decay trend, their response mismatch in the overlap region can introduce a visible discontinuity after direct fusion. This discontinuity propagates into the retrieved temperature profile, resulting in local inconsistency and degraded smoothness. Reliable full-altitude retrieval, therefore, requires not only channel overlap but also nonlinear-aware correction and retrieval-consistent fusion.

The dual-channel stitching problem is, therefore, not simply a signal-connection problem. It is a coupled problem involving detector nonlinearity, interchannel response inconsistency, and retrieval-error propagation. The goal of this study is not only to improve local smoothness in the overlap region but also to establish a physically consistent and retrieval-oriented fusion framework for dual-channel LiDAR signals. The proposed method links nonlinear detection, physics-based correction, and signal-level fusion to the final atmospheric retrieval task.

Spaceborne infrared passive remote sensing instruments, such as TIMED/SABER and Aura/MLS, provide global coverage, but their temporal resolution and field-of-view constraints limit their ability to resolve regional-scale and mesoscale

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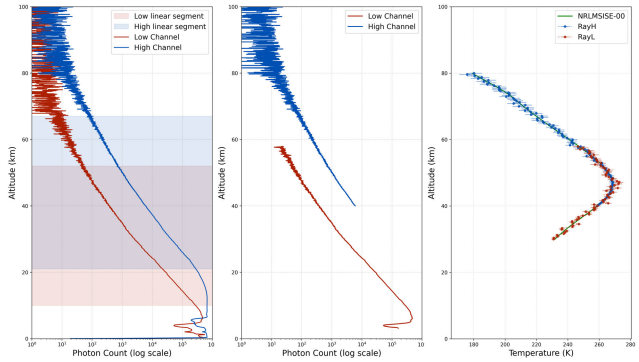


Fig. 1. Illustration of the limitations of direct dual-channel processing. (Left) Raw photon-counting profiles from the low and high channels, showing different usable altitude ranges and response characteristics. (Middle) Channelwise signal profiles used directly in the overlap region, where the low and high channels are not fully consistent in amplitude and slope. (Right) Temperature profiles independently retrieved from the two channels, showing that the signal mismatch propagates into the inversion result and leads to visible disagreement in the overlap region.

dynamic disturbances [1], [2], [3], [4], [5]. By contrast, LiDAR provides high vertical resolution, continuous nighttime operation, and adaptability to severe observational environments. It has, therefore, become one of the most effective ground-based techniques for probing temperature and density structures in the middle and upper atmosphere.

Despite these advantages, achieving high-quality and continuous full-altitude observations with ground-based LiDAR systems remains technically challenging. A core limitation arises from the restricted dynamic range of individual photomultiplier tubes (PMTs), which makes it difficult to simultaneously accommodate strong near-range echoes and weak high-altitude backscattered signals. In near-range regions, intense atmospheric returns can easily drive the detector into nonlinear or saturated regimes, leading to signal compression and distortion of the measured profiles [6], [7], [8], [9], [10], [11], [12], [13]. Conversely, in the upper atmosphere, the rapidly decaying signal places the detector in a noise-dominated regime and substantially reduces the signal-to-noise ratio. As a result, relying on a single detection channel often fails to provide continuous and reliable atmospheric profiles across the full altitude range.

More critically, photon-counting LiDAR systems inevitably suffer from nonlinear effects under high photon flux conditions, including detector saturation and pulse pile-up, which can introduce signal compression, range bias, and systematic distortions in the vertical backscatter profile [14], [15]. Recent studies have further shown that saturation-induced nonlinearities may directly propagate into subsequent retrieval steps, resulting in systematic biases in range estimation and profile reconstruction, particularly in strong near-surface return regions [16]. These issues make detector nonlinearity not merely a hardware problem but also a retrieval-level problem that directly affects the physical reliability of the final atmospheric products.

To mitigate these limitations, multichannel LiDAR architectures have been widely adopted in many studies worldwide. Typical configurations include simultaneous acquisition of analog and photon-counting signals, dual-telescope receiving structures, and multiwavelength transmission schemes. Among these designs, dual-telescope systems are particularly attractive because their strong optical separation capability enables independent optimization for low- and high-altitude signal receptions. Nevertheless, such systems still suffer from practical challenges, including inconsistent channel gains, large disparities in response dynamic range, and the lack of unified and robust data-processing strategies.

Typical systems employing Licel transient recorders to simultaneously acquire analog and photon-counting signals [17], [18], [19] can achieve dynamic ranges on the order of 10^5 by combining dead-time correction and weighted stitching techniques. In addition, systems equipped with Hamamatsu wide-dynamic-range PMT modules have demonstrated improved tolerance to strong near-surface signals while maintaining sensitivity to weak high-altitude echoes [20]. However, these hardware-level improvements alone cannot fully eliminate interchannel inconsistencies caused by detector nonlinearity and system heterogeneity.

Accordingly, considerable effort has been devoted to the development of channel stitching algorithms. Early studies investigated signal splicing in dual-receiver LiDAR systems and slope-based analog/photon-counting signal combination methods [21], [22]. Feng et al. [23] and [24] developed quadratic-polynomial (QP)-based gluing methods for analog/photon-counting signal combination, improving the stability of conversion-coefficient estimation. Other studies introduced piecewise nonlinear splicing and variance–mean consistency constraints to automatically determine fusion coefficients, ensuring Poisson compatibility between analog and photon-counting signals [25]. Liu et al. [26] further introduced a sliding-window-based (SW) gluing framework with adaptive fitting-interval selection and nonlinear correction to enhance stitching robustness under complex atmospheric conditions. Nevertheless, most existing methods still depend on predefined thresholds, local fitting assumptions, or empirically designed transition rules, which may reduce their robustness under varying system configurations, detector heterogeneity, and saturation conditions.

Existing channel-gluing approaches can be broadly grouped into linear/slope-based fitting methods, nonlinear polynomial conversion methods, and adaptive or piecewise transition-based stitching methods. To provide a representative comparison, baseline methods were selected from the latter two categories in this study, namely, a QP gluing method and an SW gluing method. These two baselines represent, respectively, a classical nonlinear conversion strategy and a recent adaptive fitting-interval selection strategy. In contrast, the proposed framework jointly considers saturation detection, nonlinear calibration, and fusion consistency before inversion, so that the objective is not only smoother local stitching but also improved consistency in subsequent density and temperature retrievals.

Recent studies have continued to emphasize the importance of robust signal correction, retrieval consistency, and improved adaptability in advanced atmospheric LiDAR and related optical sensing applications. In particular, recent work has further highlighted the value of data-driven processing

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strategies, cross-configuration robustness, and retrieval-oriented analysis in complex sensing scenarios [27], [28], [29]. Meanwhile, recent developments in signal fusion and intelligent optical information processing have also suggested that physically consistent integration across heterogeneous sensing conditions is becoming increasingly important for reliable downstream inversion and interpretation [26], [30]. These recent studies further motivate the development of the present framework, which aims to jointly address nonlinear response detection, physically constrained correction, and retrieval-consistent dual-channel fusion.

However, these studies either focus on detector-level correction without retrieval-oriented fusion or do not systematically evaluate heterogeneous PMT configurations. This motivates this work, which combines nonparalyzable physical modeling, extreme gradient boosting (XGBoost) residual correction, and retrieval-consistent dual-channel fusion to provide a unified framework for both isomorphic and heterogeneous LiDAR systems.

To further expand the effective detection dynamic range and enable simultaneous observation of strong low-altitude and weak high-altitude signals, dual-channel signal fusion has become a practical strategy. In such systems, the detection task is divided into a low-altitude channel optimized for strong returns and a high-altitude channel designed for weak echoes. During fusion, valid data pairs within the overlapping altitude range are selected, and interchannel mapping relationships are established to preserve signal continuity and physical consistency [15], [31], [32], [33], [34], [35], [36]. However, when nonlinear response, detector heterogeneity, and retrieval consistency are considered simultaneously, existing approaches still show limited automation, adaptability, and cross-system generalization capability.

Motivated by these issues, this work develops a signal fusion framework with strong cross-system adaptability for dual-channel Rayleigh LiDAR. The framework is capable of automatically identifying saturation and nonlinear response regions and performing stable nonlinear correction and channel stitching under physical constraints. It integrates multiple stages, including preprocessing, saturation detection, nonlinear calibration, signal-level fusion, physical inversion, and uncertainty evaluation, with the goal of improving the continuity of middle- and upper-atmospheric observations, enhancing the accuracy of temperature and density retrievals, and providing a robust foundation for future multiplatform and multimode LiDAR observation systems.

The novelty of this work lies in three aspects. First, an anomaly transformer-based strategy is introduced to automatically identify the onset of nonlinear response and determine the valid fusion interval, avoiding reliance on fixed empirical thresholds. Second, a physically constrained nonlinear correction framework is constructed by combining a nonparalyzable detector model with XGBoost residual learning, which improves the reconstruction accuracy of saturation-affected segments. Third, a retrieval-consistent signal-level fusion scheme is developed through interchannel nonlinear mapping and three-stage weighted stitching, and its effectiveness is verified under both isomorphic and heterogeneous PMT configurations. These improvements make the proposed method more robust for full-altitude Rayleigh LiDAR profile reconstruction and subsequent atmospheric retrieval.

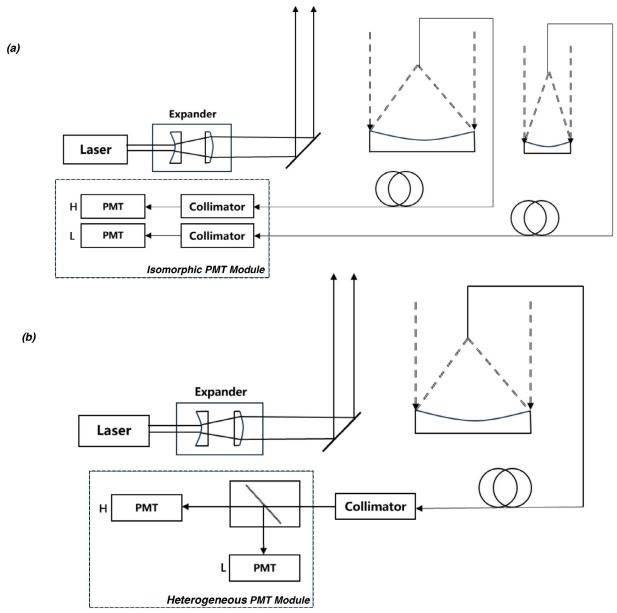


Fig. 2. Two representative LiDAR system configurations used in this study. (a) LiDAR system with an isomorphic PMT module, in which the low- and high-altitude channels are formed by two parallel receiver branches with the same collimator–PMT structure. (b) LiDAR system with a heterogeneous PMT module, in which the received signal is first coupled through a common collimator and then divided inside the PMT module into high- and low-response branches.

II. LiDAR System Configurations and Associated Challenges

A. LiDAR System Configurations and Response Discrepancies

The performance of a dual-channel LiDAR system is closely related to its optical splitting strategy, receiving structure, and detector–response characteristics. In practical observations, the received echoes usually span a very large dynamic range: strong near-range signals may drive the low-altitude channel into nonlinear compression or saturation, whereas weak high-altitude signals often suffer from low signal-to-noise ratio. Therefore, dual-channel designs are widely adopted to extend the effective detection range and to support continuous atmospheric profiling from the lower to the upper altitude regions.

In this study, the terms “isomorphic” and “heterogeneous” refer specifically to the structural form of the PMT module in the dual-channel receiving subsystem, rather than to the overall LiDAR architecture.

Fig. 2 shows two representative LiDAR system configurations considered in this study. In both cases, the received signal is divided into low- and high-altitude channels to improve the overall dynamic-range coverage and support full-altitude atmospheric detection.

To evaluate the adaptability of the proposed fusion method under different optical structures and detector configurations, two representative engineering platforms are selected for comparative analysis.

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1) *LiDAR System With an Isomorphic PMT Module:* The low- and high-altitude channels are implemented by two parallel receiver branches with the same collimator–PMT structural form. In other words, the two branches are isomorphic at the PMT-module level, while their effective response may still differ because of channel gain setting, optical coupling efficiency, geometric overlap, and alignment conditions. This configuration is used to evaluate whether the proposed fusion algorithm can adapt to interchannel discrepancies arising mainly from response-level differences under the same module structure.
2) *LiDAR System With a Heterogeneous PMT Module:* For example, the atmosphere profiling synthetic observation system (APSOS) wind–temperature detection LiDAR employs a common front-end receiving path, where the received signal is first coupled through a collimator and then divided inside the heterogeneous PMT module into two response branches, in which two different PMT models are employed: H10721 for the low-altitude RayL channel and the H7421 series for the high-altitude channel. In this case, the interchannel discrepancy is introduced not only by signal intensity but also by detector heterogeneity.

For the 532-nm Rayleigh-detection configuration considered in this study, the low-altitude channel typically covers approximately 10–40 km, where the signal is strong and a nonlinear response is more likely to occur. The high-altitude channel mainly covers about 30–80 km, where the signal is weaker, but the usable detection height is greater. Both channels use photon-counting acquisition. Because of differences in gain, saturation threshold, signal-to-noise ratio, and effective altitude range, neither channel alone can provide a complete and reliable atmospheric profile over the full observation range.

These system characteristics motivate the need for a robust fusion strategy. Through experiments on both dual-channel LiDAR platforms with isomorphic and heterogeneous PMT modules, this study evaluates whether the proposed method can maintain stable performance under different optical structures and detector configurations. The following sections, therefore, focus on nonlinear-response detection, correction, and retrieval-consistent fusion of the low- and high-altitude channels.

III. Signal Fusion Algorithm Design

A. Saturation Point Detection (AT)

To extend the effective dynamic detection range of LiDAR systems while satisfying the requirements for both strong near-range signals and weak high-altitude echoes, dual-channel signal fusion architectures are widely adopted. In such systems, the detection task is divided into low- and high-altitude channels through optical or electronic separation. The low-altitude channel employs a small splitting ratio or analog acquisition to mitigate nonlinear distortion caused by strong near-surface returns, whereas the high-altitude channel utilizes a larger splitting ratio or photon-counting detection to enhance sensitivity to weak backscattered signals. This configuration effectively extends the observation altitude range and improves the reliability of subsequent retrievals.

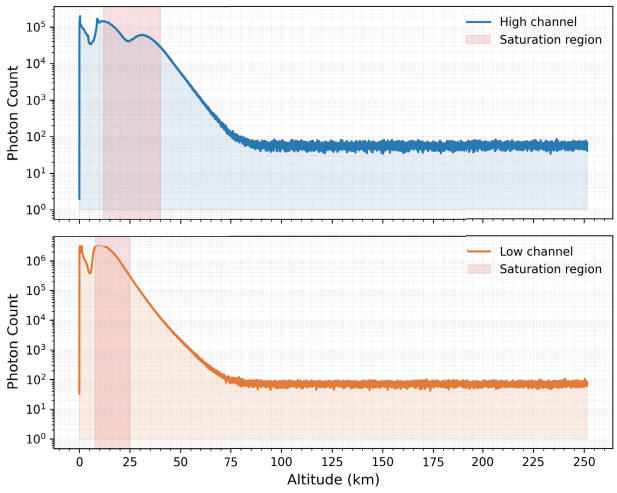


Fig. 3. Altitude-resolved photon counts showing saturation regions for the high- and low-altitude channels.

However, in practical operation, due to nonlinear detector response characteristics and the steep gradient of atmospheric signal intensity with altitude, saturation or nonlinear compression may still occur in the low-altitude channel below a certain height, while the high-altitude channel may experience count overflow or severe noise contamination in the near-range region. If such nonlinear segments are not properly identified prior to channel fusion, distorted measurements may propagate into the stitching stage and subsequent inversion processes. Therefore, accurate identification of saturation-affected regions is a prerequisite for robust signal fusion.

As illustrated in Fig. 3, the high-altitude channel exhibits poor signal quality in the lower-altitude region (approximately 10–25 km), whereas the low-altitude channel becomes noise-dominated in the upper region. Consequently, prior to channel stitching, it is essential to automatically identify saturated or distorted segments and retain only the valid signal intervals for subsequent processing.

Data-driven anomaly detection methods have recently been introduced in geoscience and remote sensing to identify structural distortions and abnormal patterns that are difficult to characterize using fixed thresholds or purely analytical models [37], [38]. Motivated by this idea, this study adopts an unsupervised anomaly transformer (AT) framework to detect nonlinear saturation regions in altitude-resolved LiDAR signals. Instead of relying on predefined thresholds, the AT uses structural association patterns along the altitude dimension to locate abrupt deviations caused by saturation or nonlinear detector response.

AT [37] is built upon a standard transformer encoder architecture and consists of three main components, as shown in Fig. 4. Compared with conventional threshold-based methods, the AT can adapt to signal patterns under varying measurement conditions, thereby reducing false detections and improving localization accuracy. First, the association modeling module employs a self-attention mechanism to capture global semantic

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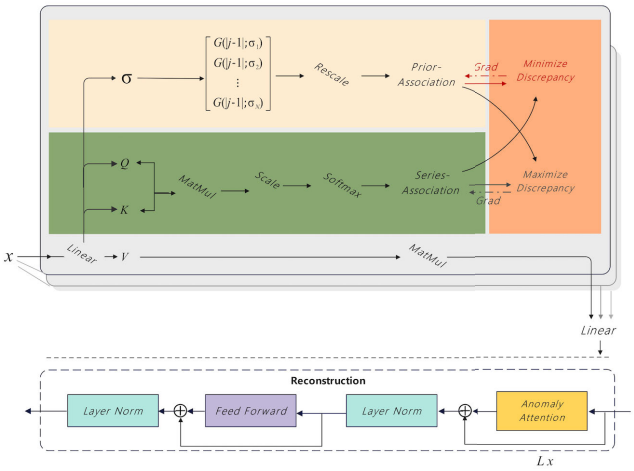


Fig. 4. Architecture of the AT.

relationships between altitude bins, constructing a series association distribution. Next, an uncertainty evaluation module introduces a Gaussian-shaped prior distribution as a reference and computes the Kullback–Leibler (KL) divergence between the learned attention distribution and the prior, quantifying the stability of the association structure. Finally, an anomaly scoring module integrates attention shifts and association uncertainty to generate an anomaly score sequence, which reflects the degree of structural deviation at each altitude.

In the low-altitude LiDAR channel, strong near-ground backscatter frequently drives the detector into a nonlinear response regime, resulting in abrupt slope changes or plateau-like segments in the photon-count profile, referred to here as saturation regions. The AT identifies these mutation points from the perspective of structural association modeling. By evaluating the attention distribution between each altitude bin and its surrounding context, and comparing it with a prior association pattern, the AT determines whether a given altitude bin exhibits structural deviation indicative of saturation onset.

Specifically, let the altitude-resolved photon-counting sequence acquired by the LiDAR be defined as

$$\mathbf{X} = \{x_i\}_{i=1}^{T} \in \mathbb{R}^T \tag{1}$$

where x_i denotes the photon count at the ith altitude bin and T represents the total number of altitude layers (e.g., covering the range of 25–70 km). The model first constructs the query (Q), key (K), and value (V) vectors through linear mappings and computes the standard self-attention weights as

$$A_{i,j} = \frac{\exp\left(\frac{Q_i \cdot K_j^{\mathrm{T}}}{\sqrt{d}}\right)}{\sum_{j=1}^{T} \exp\left(\frac{Q_i \cdot K_j^{\mathrm{T}}}{\sqrt{d}}\right)} \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{T\times T}$ is the attention matrix, which characterizes the semantic dependency between the current altitude bin and all other bins, and is referred to as the *series association* distribution. Here, $\mathbf{Q}_i$ and $\mathbf{K}_j$ denote the query and key feature vectors corresponding to the ith and jth altitude bins, respectively; d is the feature dimension of the query/key vectors in the scaled dot-product attention; and T is the total number of altitude bins.

For the prior association distribution, a Gaussian prior is constructed according to the relative distance between altitude bins. Specifically, for the ith altitude bin, the prior association weight to the jth bin is defined as

$$\tilde{P}_{i,j} = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(j-i)^2}{2\sigma_i^2}\right)$$

where σ_i is a learnable scale parameter controlling the local receptive width around the ith altitude bin. The prior association distribution is then obtained by rowwise normalization

$$P_{i,j} = \frac{\tilde{P}_{i,j}}{\sum_{k=1}^{T} \tilde{P}_{i,k}}.$$

This construction reflects the assumption that, under normal unsaturated conditions, each altitude bin is more strongly associated with its neighboring bins than with distant bins, thereby providing a physically reasonable prior for anomaly discrimination in altitude-resolved LiDAR profiles. The anomaly score at altitude bin i is then defined as the KL divergence between the learned attention distribution $\mathbf{A}_i$ and the prior distribution $\mathbf{P}_i$

$$s_i = D_{\mathrm{KL}}\left(\mathbf{A}_i \| \mathbf{P}_i\right) = \sum_{j=1}^{T} A_{i,j} \log\left(\frac{A_{i,j}}{P_{i,j}}\right) \tag{3}$$

where s_i denotes the anomaly score at the ith altitude bin, and $A_{i,j}$ and $P_{i,j}$ are the learned series association weight and the normalized prior association weight between the ith and jth altitude bins, respectively.

A sharp increase in the anomaly score indicates a structural mutation in the signal profile, suggesting a transition from the linear response regime to a nonlinear compression or saturation regime. By analyzing the anomaly score sequence and applying a threshold-based criterion, the onset and extent of saturation can be automatically determined.

To improve reproducibility, the training data and preprocessing procedure used for the AT are summarized here. The model was trained in an unsupervised manner using approximately 1200 historical low-altitude-channel photon-counting profiles collected under routine operating conditions. Before training, the input sequences were preprocessed by background subtraction, altitude registration, and normalization.

The main implementation settings of the AT used for saturation detection are summarized in Table I, including the sliding-window length, encoder depth, number of attention heads, training strategy, and discrepancy criterion. The model was trained for up to 10 epochs, and early stopping was applied when the validation discrepancy no longer improved.

Based on in situ measurements, the proposed AT framework is used to locate the onset of nonlinearity in the low-altitude channel under strong near-ground backscatter conditions. When the anomaly score exhibits a pronounced local increase that coincides with an abrupt slope change in the photon-count profile, the corresponding altitude is identified as the transition point from linear to nonlinear response. These detected saturation boundaries are subsequently used to guide nonlinear correction and to constrain the selection of valid

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TABLE I
IMPLEMENTATION DETAILS OF THE AT USED FOR SATURATION DETECTION

Item	Setting
Input dimension	1
Output dimension	1
Sliding window size	100
Embedding dimension (d_{model})	256
Number of attention heads	8
Number of encoder layers	3
Feed-forward dimension (d_{ff})	256
Optimizer	Adam
Initial learning rate	10^{-4}
Batch size	32
Training strategy	Early stopping within 10 epochs
Onset decision rule	Local maximum of anomaly score with slope mutation

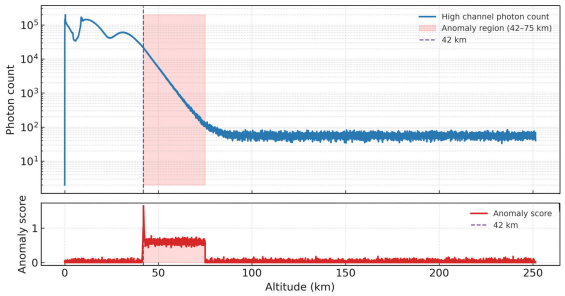


Fig. 5. Altitude-resolved photon-counting profiles of the low-altitude channel and the corresponding anomaly scores produced by the AT.

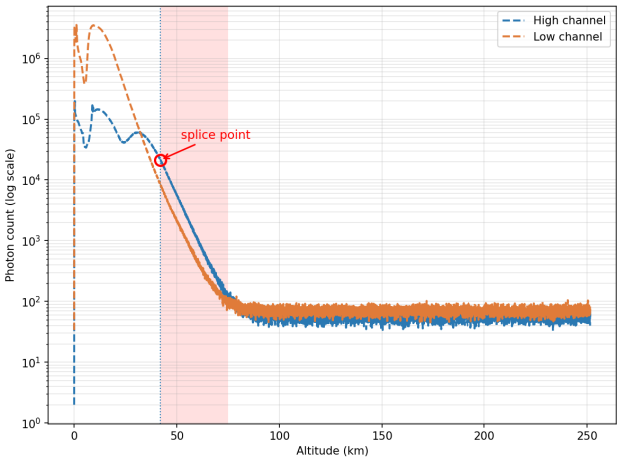


Fig. 6. High–low channel overlap and splice point selection guided by the AT.

overlap regions for channel stitching. A representative example of the AT-based saturation-boundary detection result is shown in Fig. 5, where the low-altitude-channel photon-counting profile and the corresponding anomaly-score sequence are presented.

As shown in Fig. 6, assisted by the association modeling and KL-divergence-based anomaly scoring of the AT, the altitude of 42 km is selected as the reference splice point, and the range of 42–75 km is defined as the linear overlap region. Within this region, the amplitude ratio between the high- and low-altitude channels remains stable with altitude, and the AT-derived anomaly score $s(z)$ consistently stays at a low level, indicating that the interchannel structural association is stable and suitable for scale calibration and data fusion. Beyond this range, $s(z)$ increases and is accompanied by ratio drift, suggesting the influence of field-of-view mismatch, signal-to-noise ratio degradation, or nonlinear effects; therefore, these regions are excluded from the fusion basis. The identified overlap region is subsequently used for scale estimation and channel stitching, while the AT anomaly score is continuously monitored to assess fusion quality.

B. Nonlinear Correction Based on Physical Modeling and Residual Learning

Under strong near-range echo conditions, the low-altitude photon-counting channel may enter a nonlinear compression regime before full saturation. This causes the measured count rate to deviate from the ideal linear response and becomes significant when strong backscatter exceeds the detector's effective linear dynamic range. To restore a physically meaningful photon-counting response in this region, a nonlinear correction framework is constructed by combining a nonparalyzable detector model with XGBoost residual learning.

The nonparalyzable model provides a physically interpretable baseline correction by capturing the dominant dead-time-induced compression trend of the PMT. However, practical observations are also affected by detector aging, gain fluctuation, temperature drift, and power variation, which introduce residual deviations beyond the idealized physical model. Therefore, a data-driven residual-learning strategy is further introduced to compensate for the remaining model mismatch.

The corresponding response relationship is given by

$$N_c = \frac{N_m}{1 + \tau N_m} \tag{4}$$

where N_c denotes the measured photon-counting rate, N_m represents the corrected estimate of the true photon count, and τ is the system dead time (in s), which is obtained from detector calibration under controlled photon-counting conditions and treated as a fixed detector parameter in this study.

By inverting (4), the measured photon-counting rate can be preliminarily corrected as

$$N_{\text{phys}} = \frac{N_c}{1 - N_c \tau} \tag{5}$$

where N_{phys} denotes the baseline-corrected photon-counting rate predicted by the physical model.

The residual is then defined as the difference between the reference photon-counting rate N_{true} and the physical-model output N_{phys}

$$R = N_{\text{true}} - N_{\text{phys}} \tag{6}$$

where N_{true} denotes the reference photon-counting rate used as the supervision target for XGBoost residual learning. In this study, N_{true} is not treated as an idealized theoretical value but is approximated from historical unsaturated observations collected under comparable operating conditions. These reference

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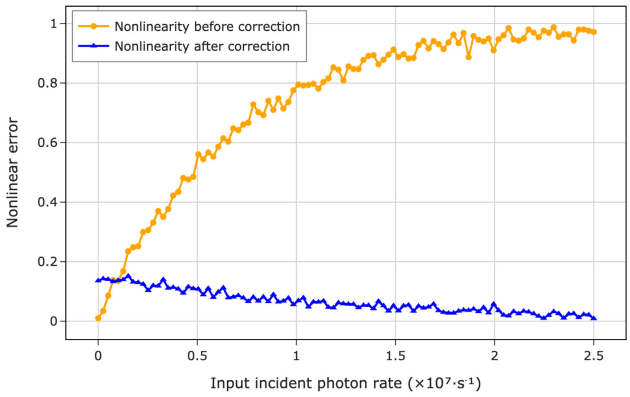


Fig. 7. PMT nonlinearity before and after correction.

profiles are selected from measurements in which the detector response remains within the normal unsaturated regime and are then aligned to the target observations through altitude registration and signal normalization.

The feature vector used for XGBoost residual learning is constructed from the measured count rate, the physical-model prediction, altitude information, and local signal-context descriptors, including the original photon-counting value, local sliding-window statistics, gradient-related features, and background-noise estimation. The regression model is trained to predict the residual term R, and the final corrected photon-counting rate is obtained by

$$N_{\text{corr}} = N_{\text{phys}} + \hat{R} \tag{7}$$

where $\hat{R}$ denotes the residual estimated by XGBoost. In this way, the proposed framework preserves the physical interpretability of the dead-time model while improving correction accuracy through data-driven compensation.

The nonlinear error curves of the PMT under different incident photon flux levels are shown in Fig. 7. The curve labeled "Before correction" represents the raw unmodeled response error, while the "After correction" curve corresponds to the result obtained using the nonparalyzable system model. It is evident that the traditional model effectively mitigates the rapid growth of nonlinear errors. However, some residual fluctuations remain in the high-intensity region, which motivates the introduction of residual learning for further compensation.

To better visualize the structure of the proposed method, Fig. 8 shows the overall correction process in a modular form, including physical-model estimation, residual construction, feature extraction, and XGBoost prediction.

The changes in photon count over the altitude range of 42–80 km before and after saturation correction are shown in Fig. 9. In the magnified view of the 45–50-km range, the uncorrected data shows a significant downward slope, while the corrected result restores a near-monotonic decay that aligns more closely with the physical process and provides more reliable input for subsequent stitching and temperature inversion. After joint correction using the nonparalyzable model and XGBoost residual learning, the signal recovers the expected monotonic decay characteristic, demonstrating that the proposed method effectively suppresses the systematic bias caused by pulse pile-up.

In summary, the proposed nonlinear correction framework combines the physical interpretability of the nonparalyzable detector model with the flexibility of XGBoost residual learning. The corrected signal restores a more physically consistent monotonic decay in the nonlinear segment and provides more reliable input for subsequent channel mapping, stitching, and temperature inversion.

C. Channel Mapping Modeling and Weighted Stitching

After nonlinear correction of the low-altitude channel, a fusion model is constructed to combine the low- and high-altitude channels for continuous LiDAR detection over the full altitude range. A single channel cannot provide reliable measurements across all altitudes because strong near-range signals may saturate the low-altitude channel, whereas weak high-altitude returns require enhanced sensitivity. Therefore, physically consistent channel fusion is essential for ensuring continuity of the stitched signal and reliability of the subsequent inversion results.

1) Fitting and Modeling Method for the Fusion Segment: The first step in the fusion process is to establish the mapping relationship between the low- and high-altitude channels. In this study, the nonlinear correction model is proposed for the segment between the saturation altitude z_{sat} and the stable region of the atmospheric model $z_{\text{ref}} = 70\,\text{km}$. This segment has a good signal-to-noise ratio and smoothness. Traditional fusion methods usually require several empirical steps and may be less stable in abrupt-transition regions or weak-signal sections.

To improve the model's adaptability, this study proposes using XGBoost to model the low- and high-altitude channels and correct their respective nonlinearities (such as smoothness, local gradients, and background noise) as features for training the model. The high-altitude signal correction model can be expressed as

$$\hat{P}_H(z) = f_{\text{XGB}}(x(z)) \tag{8}$$

where $\mathbf{x}(z_i) \triangleq \mathbf{x}_i$ denotes the input feature vector constructed at the ith altitude bin z_i, which includes the signal intensity and its spatial–structural descriptors (e.g., local moving average, gradient/slope, signal-to-noise ratio, or estimated background noise). The channel-mapping model is trained in a supervised manner using historical observations to learn the nonlinear response relationship between the low- and high-altitude channels. To avoid overly optimistic performance caused by information overlap between adjacent altitude bins introduced by sliding-window features, the training/validation/test sets are partitioned using a mutually exclusive, nightwise (or profilewise) grouping strategy: all altitude bins from the same night are assigned to a single split, and the test set consists of independent nights not involved in model training. This design enables a more objective evaluation of the model's predictive stability on unseen observation batches and improves mapping accuracy under nonstationary conditions such as cloud perturbations and abrupt boundary-layer transitions.

It is important to note that the XGBoost algorithm used in this study not only corrects for nonlinearities in previous sections but also mimics the physical model's residuals, enhancing the model's flexibility and adaptability to different

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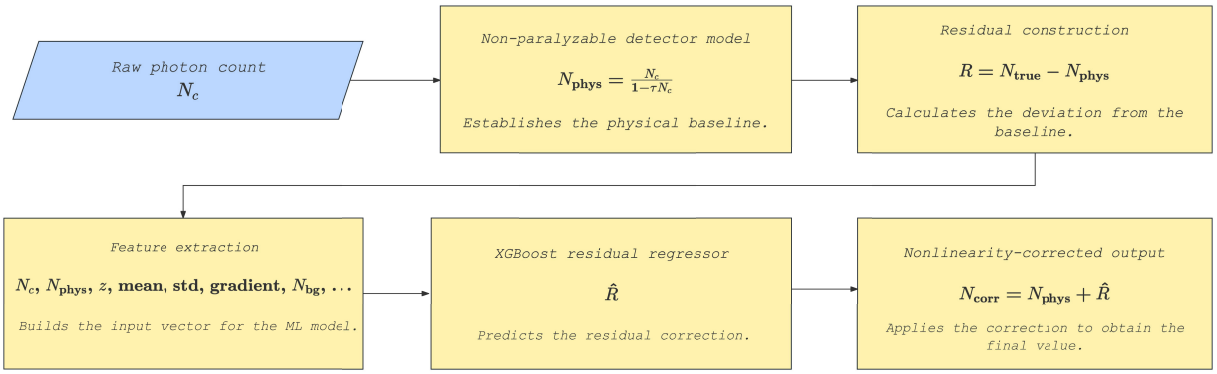


Fig. 8. Residual-learning pipeline for photon-counting nonlinearity correction based on the nonparalyzable detector model and XGBoost residual regression.

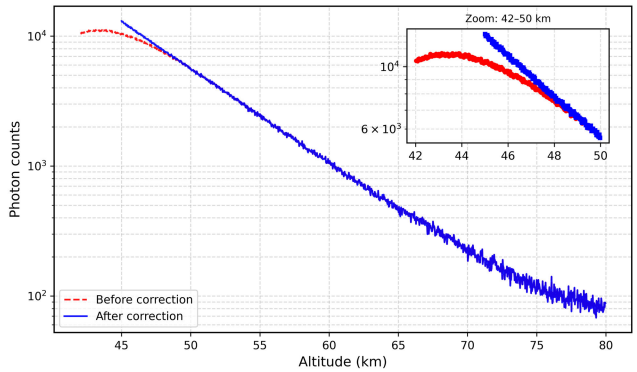


Fig. 9. Photon counts before and after nonlinear correction.

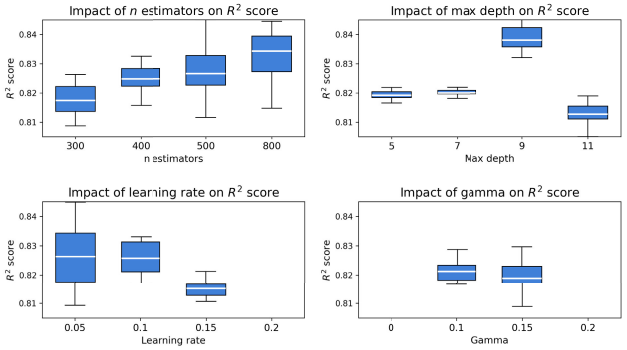


Fig. 10. Impact of XGBoost hyperparameters on R^2 for the channel-mapping regression.

tasks. At this stage, the role of XGBoost is to learn the inter-channel response relationship from data and to improve the adaptability of the mapping model under varying observational conditions.

2) Hyperparameter Tuning and Performance Sensitivity Analysis: The regression prediction results of the optimized PCA dimensionality reduction and XGBoost model are shown in Fig. 10. The improved model shows stable fitting performance on both the training and test sets, with $R^2 = 0.84$, RMSE = 0.45, and $R_P = 3.18$. As the number of $n_{\text{estimators}}$ increases from 300 to 800, the model's fitting ability improves, with R^2 rising from 0.82 to 0.84. Although increasing the tree depth (max_depth) from 5 to 11 results in only a small change in R^2, which remains between 0.81 and 0.84, the learning rate (learning_rate) and gamma have a significant impact on model performance. A lower learning rate (0.05) resulted in the highest R^2 (approximately 0.84). Compared to the model before optimization, the prediction accuracy has significantly improved, demonstrating that the proposed optimization method not only enhances the model's fit to the training data but also strengthens its predictive capability for new data. This result validates the effectiveness of the optimization strategy.

3) Signal Stitching Function and Reweighting Strategy: After establishing the interchannel response relationship, a continuous fused signal is constructed from the low- and high-altitude channels. The fusion function is designed as a three-segment stitching function

$$P_{\text{merged}}(z) = \begin{cases} P_L(z), & z < z_{\text{merge}} \\ (1 - w(z)) \cdot P_L(z) & \\ \quad + w(z) \cdot \hat{P}_H(z), & z_{\text{merge}} \le z \le z_{\text{ref}} \\ \hat{P}_H(z), & z > z_{\text{ref}}. \end{cases} \tag{9}$$

Here, the stitching starting point z_{merge} can be consistent or adjusted based on actual residual corrections. The weight function $w(z)$ is defined as

$$w(z) = \frac{z - z_{\text{merge}}}{z_{\text{ref}} - z_{\text{merge}}}. \tag{10}$$

This weight function not only ensures the smooth transition between two channels but also accomplishes the adjustment based on the fusion of the signal strength and the continuity of the signal structure. The weight function is a linear form that ensures smooth fusion across the entire range.

This linear weighting strategy realizes the transition between the low- and high-altitude channels. By using XGBoost, we model the signal's residual dynamic adjustment to the reweighting factor and implement a soft fusion of the two signals.

The final step after completing the nonlinear LiDAR signal fusion and stitching is to perform a comprehensive performance evaluation across the full range. This assessment evaluates the model's fitting accuracy using the training set, as well as its robustness against various data variations. Through this process, we can further validate that the model is stable under different conditions, ensuring that the model's predictions remain accurate across new datasets.

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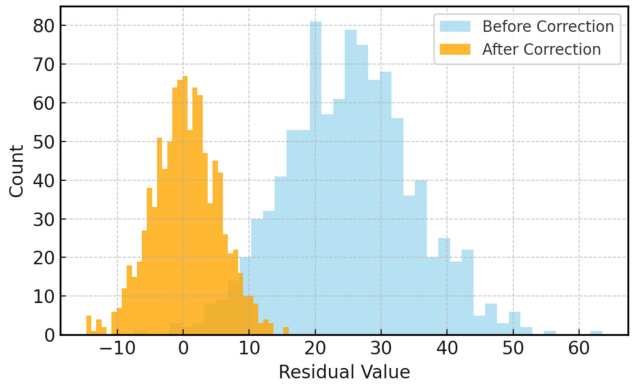


Fig. 11. Residual distributions before and after nonlinear-response correction.

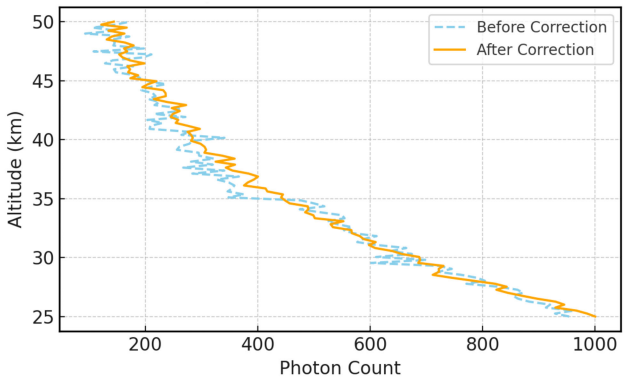


Fig. 12. Signal profiles (25–50 km): precorrection/postcorrection comparison.

D. Model Performance Evaluation and Validation

To evaluate the stability and generalizability of the nonlinear-response correction and interchannel mapping components of the proposed fusion algorithm in a multiplatform setting, we conducted a systematic assessment of two XGBoost-based models. The evaluation covers residual-fitting accuracy, training convergence behavior, feature-contribution structure, and cross-environment validation performance, with the aim of examining the models' robustness, interpretability, and applicability in real LiDAR observation scenarios.

The two XGBoost models are designed for: 1) residual regression for PMT nonlinearity correction and 2) low-to-high channel mapping. Because the input features include statistics computed with sliding windows, randomly splitting the dataset at the altitude-bin level would create overlapping windows between adjacent bins, leading to information leakage and overestimated performance. To mitigate this issue, we adopt a profilewise/nightwise group split strategy: all altitude bins from the same profile are assigned exclusively to a single subset, and the training, validation, and test sets are mutually exclusive at the profile/night level. For cross-platform validation, we further employ a leave-one-platform-out protocol to assess the transferability of the models.

1) Residual Fusion Performance and Signal Consistency: In the nonlinear correction task for PMT, the constructed model predicts the residual error between the predicted values and the actual measured photon counts as the training objective, combining with the sliding mean, gradients, and other multidimensional features of the signals. Fig. 11 illustrates the model's residual distribution in the typical nonlinear compression region (25–45 km). Before applying the AI correction, the residual shows significant bias in the original physical model's output, revealing difficulties in adjusting the model's response to the LiDAR system's nonlinearities. After the XGBoost model correction, the residual distribution becomes close to a normal distribution, and the standard error decreases significantly, demonstrating the strength of the nonlinear correction.

Fig. 12 compares the signal profiles before and after correction. The corrected signal restores continuity near slope-transition and plateau-like structures, resulting in a smoother overall profile. This improvement provides a more reliable input for subsequent stitching and temperature inversion.

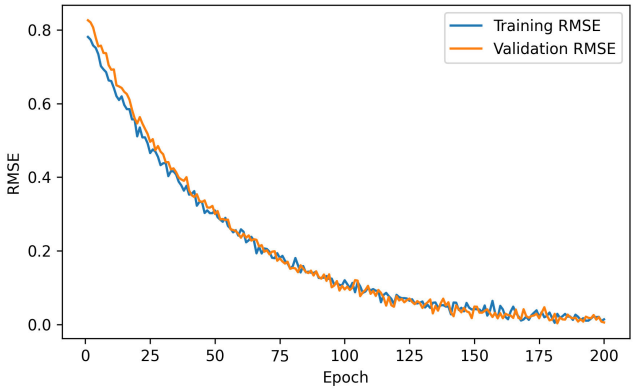


Fig. 13. Training/validation RMSE convergence of the XGBoost residual model.

2) Model Training Process and Prediction Performance Analysis: Fig. 13 shows the convergence curves of the XGBoost model, where the root-mean-square error (RMSE) on the training and validation sets is plotted as a function of boosting iterations. The RMSE decreases rapidly during the early stage (approximately the first 200 iterations) and then gradually levels off. The training and validation curves remain close to each other without a persistent divergence, indicating that, under the current data-splitting strategy, the model does not exhibit pronounced overfitting and maintains a stable prediction error on the validation set. Combined with the mutually exclusive profilewise/nightwise group split adopted in this study, these results more reliably reflect the model's stability on unseen observation batches that were not involved in training.

It can be observed that the RMSE on the training and validation sets gradually converges as the number of boosting iterations increases, while the gap between the two curves remains small. This indicates that the error levels on the training and validation data are largely consistent and that the fitting process is stable. It should be noted, however, that the closeness of the curves primarily reflects stability under the validation-set distribution.

3) Feature Contribution and Model Interpretation Analysis: To enhance the model's interpretability, this study further analyzes the contribution of each feature in the model using the gain metric. Fig. 14 shows the ranking of feature importance according to gain. The figure demonstrates that, among the

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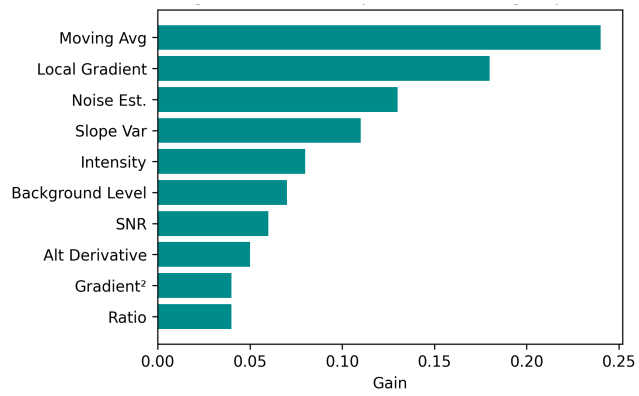


Fig. 14. Top-ten feature importance by gain (XGBoost).

features in the training model, the ones with the highest importance include the moving average, local gradient, noise estimation, and background level, which significantly contribute to the model's training. This ranking result, driven by data, indicates the nonlinear response between these features and the model's output.

In summary, the model trained using XGBoost shows excellent residual fitting performance, and the training model is highly consistent with the test set performance. The model is not dependent on specific configurations of equipment but still achieves efficient nonlinear fitting. The results confirm that the model, through the fusion of LiDAR and other physical signals, can provide a solid foundation for the practical application of remote sensing.

IV. Evaluation of Stitching Algorithm and Result Comparison

A. Method Overview (Density and Temperature Inversion)

After the stitching of the LiDAR signals, this study uses the Chanin–Hauchecorne (CH) method for performing temperature inversion. The temperature inversion is based on the relationship between LiDAR scattering strength and the particle number density. The process includes filtering high-frequency noise, removing background noise, and applying height correction, which restores the continuity of the signal. The formula for this calculation is

$$n(z) = n(z_0) \cdot \frac{P(z)}{P(z_0)} \tag{11}$$

where $n(z)$ represents the particle number density at height z, $P(z)$ is the scattering strength at that height, and z_0 is the reference height. During temperature inversion, based on static equilibrium and the ideal gas state equation, the temperature gradient is derived using the CH framework as

$$T(z) = -\frac{Mg}{R}\left(\frac{d\ln n(z)}{dz}\right)^{-1} \tag{12}$$

where M is the molar mass of the gas, g is the acceleration due to gravity, and R is the gas constant. This formula is used to model the temperature profile, and by considering the data in the boundary layer, it helps establish a reliable model for temperature inversion, ensuring the physical accuracy of the results.

B. Multisource Stitching and Weighted Fusion

For multisource fusion, an uncertainty-weighted L2 stitching method is used. The physical quantity at height z is obtained from two sources, denoted by $X_1(z)$ and $X_2(z)$, with corresponding uncertainties $\Delta X_1(z)$ and $\Delta X_2(z)$. The fusion estimate is calculated as

$$X_{\text{fusion}}(z) = \frac{\Delta X_2^2}{\Delta X_1^2 + \Delta X_2^2} \cdot X_1(z) + \frac{\Delta X_1^2}{\Delta X_1^2 + \Delta X_2^2} \cdot X_2(z). \tag{13}$$

The uncertainty of the fused value is

$$\Delta X_{\text{fusion}}(z) = \sqrt{\frac{\Delta X_1^2 \cdot \Delta X_2^2}{\Delta X_1^2 + \Delta X_2^2}}. \tag{14}$$

This weighting strategy highlights the contribution of the higher-quality data in the stitching region, improving the smoothness and continuity of the profile. It also effectively reduces the cumulative error from single-source measurements, resulting in a fusion that better reflects the physical principles and provides better statistical stability.

This study does not attempt to establish a full end-to-end closed-form uncertainty propagation framework for the entire processing chain. Instead, uncertainty is discussed at the signal-fusion level, and the stability of the final retrievals is evaluated empirically through temperature deviations, RMSE, and comparison with the reference atmospheric model.

C. Error Bars and Uncertainty Evaluation

To quantify the confidence of the poststitching fusion results, this study introduces the calculation of the error bars based on the statistics of the photon count. For photon-counting LiDAR, the echo statistics are primarily governed by photon-counting noise and can be approximately described by Poisson statistics. For sufficiently large photon counts, a Gaussian approximation may be used for practical uncertainty visualization. The uncertainty in the temperature or scattering inversion can be approximated as

$$\Delta T(z) \propto \frac{1}{\sqrt{N(z)}} \tag{15}$$

where $N(z)$ is the photon count at height z. As can be seen, in regions with low signal-to-noise ratios, the photon count is low, leading to large uncertainties, whereas, in high-altitude regions where the signal is weaker, the uncertainty increases. In the fusion region, the multisource fusion significantly reduces the error bars, making the error bars shorter and the profile smoother. This further leads to a uniform correction of the temperature inversion errors

$$\Delta T_{\text{norm}}(z) = \frac{T_{\text{obs}}(z) - T_{\text{model}}(z)}{T_{\text{model}}(z)}. \tag{16}$$

This is used to compare observational and model-based data, thus quantifying the systematic bias caused by model errors. These results indicate that the fusion strategy improves profile stability and provides better agreement with the reference model.

In this study, uncertainty handling is performed at two levels. At the signal level, the uncertainty of the fused profile

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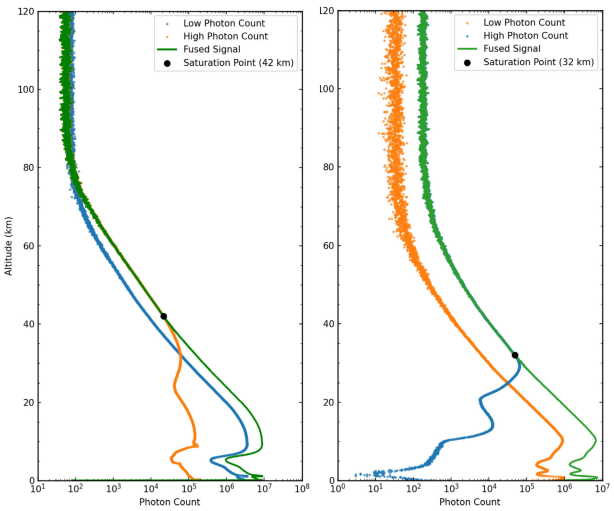


Fig. 15. Photon-count profiles (log scale) from the low- and high-channel measurements and the fused profile as a function of altitude. (Left) Isomorphic PMT system. (Right) Heterogeneous PMT system. The black marker indicates the automatically detected saturation/anomalous altitude (42 and 32 km), which is used as the transition reference for scale matching in the overlap region and for merging the two channels into a continuous profile.

is estimated through weighted variance propagation from the two input channels, which preserves the dominant contribution of photon-counting noise in the overlap region. At the retrieval level, the uncertainty of the final atmospheric products is evaluated empirically through temperature-profile deviation statistics, RMSE within the merging band, and comparison with the reference atmospheric model. This treatment is adopted because strict end-to-end analytical uncertainty propagation through nonlinear correction, residual learning, weighted fusion, and derivative-based inversion is difficult to express in a compact closed form. Therefore, the current framework combines signal-domain uncertainty propagation with retrieval-domain consistency evaluation to characterize the reliability of the final temperature and density products.

It should be noted that the signal-fusion step itself is a deterministic postprocessing operation on the original photon-counting data and does not introduce an additional independent physical noise source. Therefore, the uncertainty of the final retrievals mainly originates from the original photon-counting noise, residual errors from nonlinear correction, and error propagation in the subsequent inversion. In this study, the reliability of the final retrievals is assessed mainly through temperature deviations, RMSE, and comparison with the reference atmospheric model, rather than through a full end-to-end closed-form uncertainty derivation.

D. Comparison of Density and Temperature Inversion Between Isomorphic PMT and Heterogeneous PMT Systems

1) Comparison of L0-Level Fusion (Gluing) Profiles for Isomorphic and Heterogeneous PMT Systems and Representative Baseline Comparison: Fig. 15 presents the L0-level signal fusion results for the isomorphic and heterogeneous PMT systems. The blue and orange scatter points denote the original photon-counting profiles from the two channels, and the green curve represents the L0 fused signal obtained after scale matching in the overlap region and channel merging. The black markers indicate the automatically identified saturation/anomalous altitudes, which serve as reference points for the fusion transition.

For the isomorphic PMT system (see Fig. 15 left), the two channels exhibit good consistency in both magnitude and trend within the overlapping altitude range (approximately 30–50 km), enabling reliable estimation of the scaling factor and amplitude calibration. Based on the automatically identified saturation/anomalous altitude (around 42 km), a trustworthy overlapping region in its vicinity is selected for scale matching. The fused profile is then constructed by combining the more reliable channel segments below and above the reference point, resulting in a continuous and smooth L0-level fused profile (green curve) that preserves a physically reasonable monotonic decay over the entire altitude range.

For the heterogeneous PMT system (see Fig. 15, right), the differences in detector response and dynamic range are more pronounced, leading to larger discrepancies in signal magnitude and decay slope between the two channels in the lower and overlapping altitude ranges. Direct gluing in this case would introduce significant systematic bias. By automatically identifying the saturation/anomalous altitude and selecting approximately 32 km as the fusion reference point, scale matching is performed within the nearby overlapping region, followed by segmented channel merging. This procedure yields a smooth and continuous L0-level fused signal (green curve) across the full altitude range, effectively extending the usable dynamic range of the system and suppressing fusion-induced discontinuities.

To further evaluate the practical advantage of the proposed framework, Fig. 16 compares two representative baseline gluing methods with the proposed method under both the isomorphic and heterogeneous PMT configurations. Specifically, a QP gluing method and an SW gluing method are selected to represent a classical nonlinear conversion strategy and a recent adaptive fitting-interval selection strategy, respectively. For clarity, the comparison is organized into six subfigures: Fig. 16(a1)–(a3) corresponds to the isomorphic PMT system, and Fig. 16(b1)–(b3) corresponds to the heterogeneous PMT system. Among them, Fig. 16(a1) and (b1) shows the full-altitude fused profiles, Fig. 16(a2) and (b2) shows enlarged views of the linear region, and Fig. (a3) and (b3) shows enlarged views of the nonlinear region.

For the isomorphic PMT system, Fig. 16(a1)–(a3) shows that all three methods produce globally continuous fused profiles because the two channels already exhibit relatively good agreement in the overlap region. In particular, Fig. 16(a1) shows that the fused profiles obtained by the QP method, the SW method, and the proposed framework remain broadly consistent over the full altitude range. However, Fig. 16(a2) shows that the baseline methods still exhibit more noticeable local slope and amplitude mismatch in the linear region, whereas the proposed framework yields a smoother and more stable result. Fig. 16(a3) further shows that the proposed framework provides better consistency in the nonlinear region.

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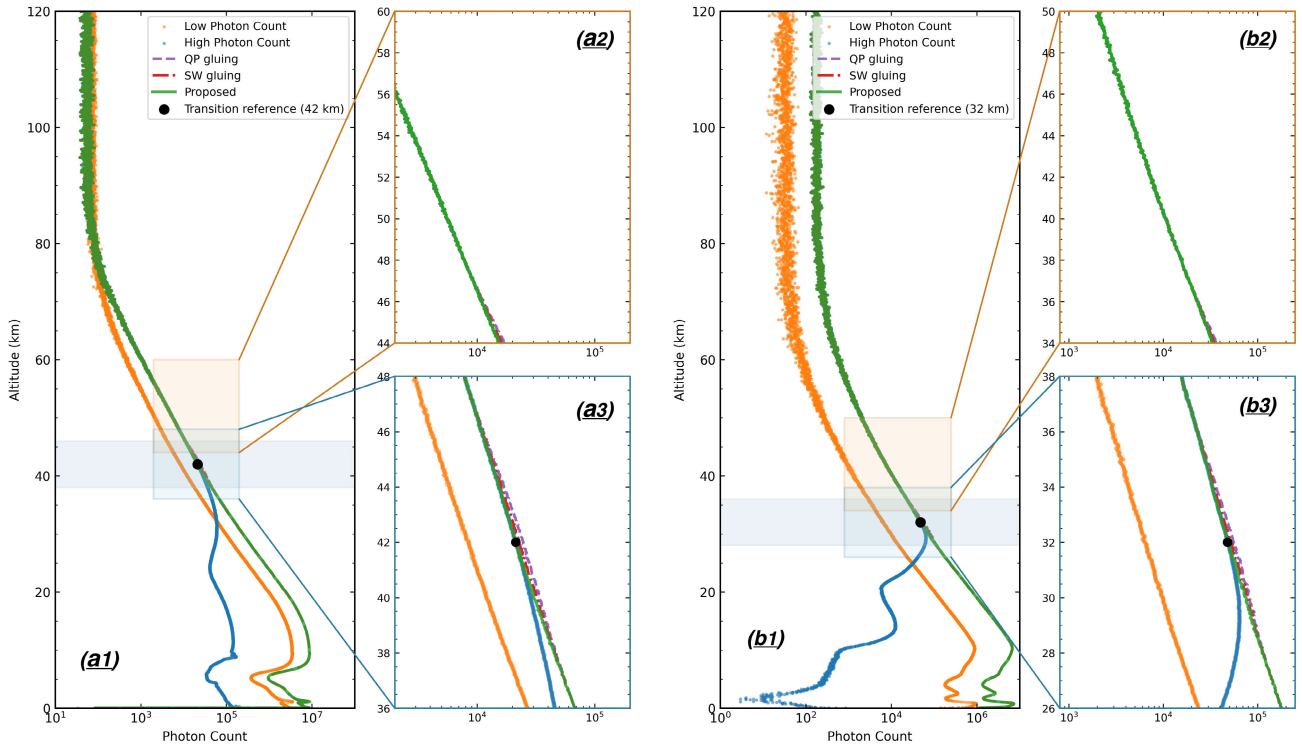


Fig. 16. Comparison of representative gluing methods and the proposed framework for the isomorphic and heterogeneous PMT systems. (a1) Full-altitude fused profiles for the isomorphic PMT system. (a2) Enlarged comparison in the linear region for the isomorphic PMT system. (a3) Enlarged comparison in the nonlinear region for the isomorphic PMT system. (b1) Full-altitude fused profiles for the heterogeneous PMT system. (b2) Enlarged comparison in the linear region for the heterogeneous PMT system. (b3) Enlarged comparison in the nonlinear region for the heterogeneous PMT system. (a) Isomorphic PMT system. (b) Heterogeneous PMT system.

These results indicate that, even when the detector responses are structurally similar, the proposed method can still improve local interchannel consistency in both the linear and nonlinear regions.

For the heterogeneous PMT system, Fig. 16(b1)–(b3) shows that the gluing task becomes more challenging because the detector–response discrepancy is more pronounced. As shown in Fig. 16(b1), the differences among methods become more evident over the full-altitude fused profiles under detector heterogeneity. In Fig. 16(b2), the baseline methods are more sensitive to fitting-interval selection and local interchannel inconsistency in the linear region, leading to larger mismatch around the fusion region. By contrast, the proposed framework maintains a smoother and more stable result. In Fig. 16(b3), the proposed framework also preserves better consistency in the nonlinear region, whereas the baseline methods exhibit relatively stronger residual inconsistency. These observations indicate that the proposed framework has stronger robustness under detector heterogeneity and nonlinear-response differences.

2) Comparison of Atmospheric Density Retrievals for Isomorphic and Heterogeneous PMT Systems: For the isomorphic PMT system, two representative retrieval pathways are compared in this study: retrieval after L0-level fusion and L2-level fusion after independent L0 retrievals, as shown in Fig. 17.

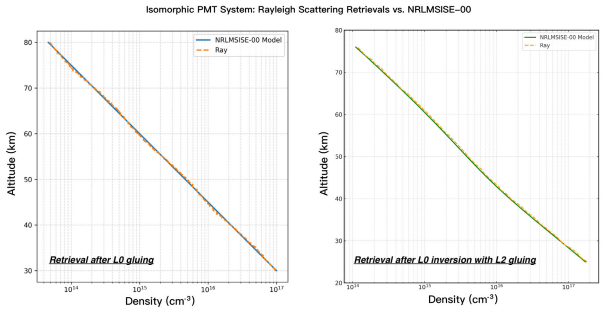


Fig. 17. Isomorphic PMT system: Rayleigh-scattering density retrievals versus NRLMSISE-00.

First, the retrieval-after-L0-fusion approach (orange curve) enforces amplitude continuity and physical consistency between the low- and high-altitude signals during the fusion stage, resulting in an overall smooth input profile for the retrieval. In the altitude range of 30–70 km, the retrieved density profile shows close agreement with the NRLMSISE-00 model (blue curve), accurately reproducing the monotonic decrease of atmospheric density with altitude. No evident discontinuities or distortions are observed in the fusion region. This indicates that performing fusion prior to retrieval effectively suppresses error propagation at the source, leading to more stable and physically consistent retrieval results. The corresponding Rayleigh-scattering density retrieval results for the heterogeneous PMT system are shown in Fig. 18.

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In contrast, the L2-level fusion after the independent L0 retrievals approach (red curve) first performs density retrievals separately along the L0 and L2 processing paths and subsequently merges the results in the overlapping region. Owing to residual differences in amplitude trends between the low- and high-altitude channels during independent retrieval, localized inconsistencies arise in the overlapping region (30–40 km), with small fluctuations appearing near the fusion point. Although the overall profile still follows the general decay trend predicted by the atmospheric model, its local smoothness is degraded. This "retrieve–then–fuse" strategy tends to amplify interchannel response differences, thereby introducing additional discontinuities in the fusion region.

Overall, retrieval after L0-level fusion outperforms L2-level fusion after retrieval in terms of profile continuity, smoothness, and error control. The former yields density profiles that are more consistent with the reference model and exhibit lower uncertainty, whereas the latter shows localized deviations near the fusion region despite preserving a reasonable global trend. These results demonstrate that, for isomorphic PMT systems, the "fuse–then–retrieve" processing pathway is more favorable for obtaining high-accuracy and physically stable atmospheric density profiles.

For the heterogeneous PMT system, two processing pathways are further evaluated: retrieval after L0-level fusion and L2-level fusion after independent L0 retrievals.

For the retrieval-after-L0-fusion scheme (orange curve), signal-level fusion is first applied to ensure the smoothness and amplitude consistency of the input profile, followed by atmospheric density retrieval. The results show that, within the altitude range of 30–70 km, the retrieved density profile exhibits close agreement with the NRLMSISE-00 model (blue curve). No visible discontinuities are observed near the fusion region, and only sparse, minor deviations are present. This indicates that, under heterogeneous system conditions, the proposed workflow effectively suppresses the propagation of interchannel differences, thereby preserving profile smoothness and physical consistency.

In contrast, the L2-level fusion after the independent L0 retrievals approach (red curve) exhibits larger amplitude discrepancies in the overlapping region of 40–50 km, with insufficient continuity near the fusion point. Due to differences in amplitude and slope responses during the independent retrieval stage, more pronounced local fluctuations and amplitude misalignments occur within the fusion band around 40–50 km. Consequently, the continuity and smoothness near the fusion point are inferior to those achieved by the retrieval-after-L0-fusion approach. Although the overall decay trend with altitude remains consistent with the reference model, this processing pathway is more prone to amplifying interchannel inconsistencies under heterogeneous conditions, resulting in inferior error control compared to L0 fusion followed by retrieval.

Overall, retrieval after L0-level fusion likewise demonstrates superior smoothness and accuracy for heterogeneous PMT systems, maintaining good agreement with the reference model despite cross-detector differences. In contrast, L2-level fusion after retrieval is more susceptible to local amplification of

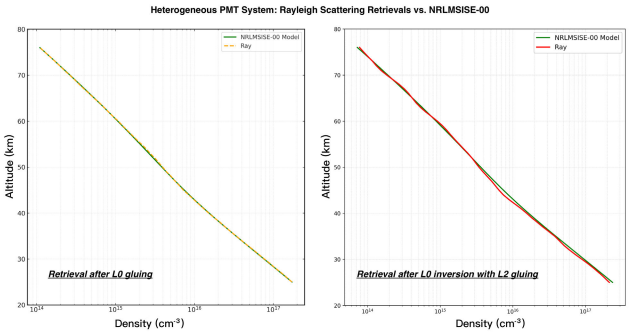


Fig. 18. Heterogeneous PMT system: Rayleigh-scattering density retrievals versus NRLMSISE-00.

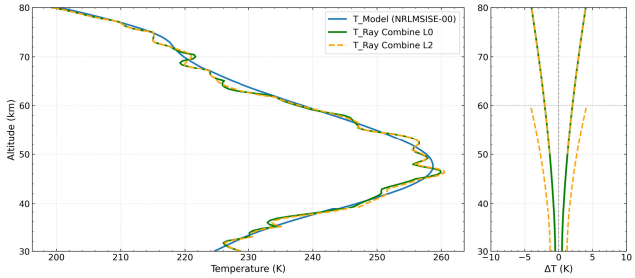


Fig. 19. Isomorphic PMT system: temperature retrieval results for the L0 (fuse–then–retrieve) and L2 (retrieve–then–fuse) workflows, compared with the NRLMSISE-00 model. The right figure shows the vertical distribution of ΔT.

interchannel discrepancies, leading to increased local discontinuities and deviations.

3) Comparison of Atmospheric Temperature Retrievals for Isomorphic and Heterogeneous PMT Systems: After completing the dual-channel signal fusion, atmospheric density retrieval based on Rayleigh scattering is first performed, followed by temperature retrieval using the CH method. The temperature from the NRLMSISE-00 model is adopted as the reference, and the temperature deviation is defined as ΔT, which is used to evaluate the impact of different fusion strategies on temperature profile continuity and error propagation.

The temperature results obtained using L0-level fusion followed by retrieval are shown in Fig. 19. The retrieved temperature profile is consistent with the model temperature over 30–80 km, reproducing the atmospheric stratification structure and showing stable performance in the middle-altitude region. The vertical distribution of ΔT indicates generally small deviations and good profile smoothness, with a maximum deviation of approximately 3.7 K. Quantitative analysis further shows that the temperature RMSE within the fusion region is 1.62 K, demonstrating that L0-level fusion constrains interchannel amplitude continuity at the signal level and suppresses error accumulation during retrieval.

The temperature results derived from L2-level fusion after independent retrievals are also shown in Fig. 19. The independently retrieved temperatures from the low- and high-altitude channels are generally consistent within 30–60 km; however, slight misalignments and localized fluctuations remain in the fusion-related layers of approximately 50–60 km. This behavior indicates that the "retrieve–then–fuse" strategy is more susceptible to propagating interchannel response differences

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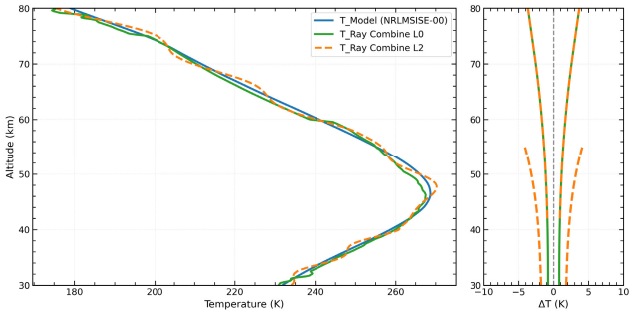


Fig. 20. Heterogeneous PMT system: temperature retrieval results obtained using both the L0 (fuse–then–retrieve) and L2 (retrieve–then–fuse) workflows, compared with the NRLMSISE-00 model. The right figure shows the vertical distribution of ΔT.

and noise into the fusion region. The ΔT distribution shows that deviations at most altitudes are confined within ± 4 K, while larger deviations occur at higher altitudes, with maximum values of approximately 4.1–4.3 K. In the overall statistical analysis, the maximum deviation in the fusion region for the L2 approach reaches 4.8 K, with a corresponding RMSE of 2.45 K, indicating lower smoothness and consistency than the L0 approach.

Overall, both processing pathways are capable of reproducing the vertical variation of atmospheric temperature over the altitude range of 30–80 km with good agreement with the reference model, demonstrating their general applicability. However, in terms of fusion-region continuity, suppression of local fluctuations, and control of error propagation, the L0-level fusion followed by the retrieval approach exhibits clear advantages, yielding smoother temperature profiles, smaller deviations, and more favorable quantitative error metrics.

Based on the comparative results obtained for the isomorphic PMT system, the following section further presents temperature retrievals for the heterogeneous PMT system under both L0 and L2 processing pathways and discusses the characteristics and impacts of fusion errors under enhanced detector–response differences.

For the heterogeneous PMT system, Fig. 20 presents the temperature retrieval results obtained using both the L0 (fuse–then–retrieve) and L2 (retrieve–then–fuse) pathways. The temperature profile retrieved by the L0 approach (orange curve) agrees well with the NRLMSISE-00 model (green curve) over the altitude range of 30–80 km and successfully reproduces the stratified structure of atmospheric temperature variation with altitude. The corresponding vertical distribution of the temperature deviation ΔT (defined as the difference between the retrieved temperature and the model temperature) indicates that deviations at most altitudes are confined within ± 3 K, with a maximum deviation of approximately 3.8 K. These results indicate that, under heterogeneous detector conditions, L0-level fusion preserves good interchannel continuity at the signal level and provides a physically consistent basis for subsequent temperature retrieval.

By comparison, the L2 approach also captures the overall temperature variation trend across the full altitude range, but slight misalignments and localized fluctuations remain evident in the fusion-sensitive layer around 45–55 km. This behavior indicates that the retrieve–then–fuse workflow is more likely to propagate interchannel response differences and noise characteristics into the fusion region, thereby reducing local smoothness. As shown in the right figure of Fig. 20, for most altitudes, $|\Delta T|$ remains within 4 K, whereas the deviation becomes relatively larger at higher altitudes (approximately 75–80 km), reaching a maximum of about 4.2 K. Overall, although the L2 pathway remains usable for capturing the global temperature variation trend, its local consistency and continuity in the fusion region are inferior to those of the L0 pathway.

Considering the results for both isomorphic and heterogeneous systems, the original observations generally suffer from saturation or nonlinear effects induced by strong low-altitude returns, as well as the inherent limitation of noise-dominated weak signals at higher altitudes. Under these conditions, the L0 fusion strategy shows better robustness for both system types by providing a smoother signal-level transition and reducing the propagation of fusion-induced errors into subsequent density and temperature retrievals. As a result, the retrieved profiles are more continuous, stable, and physically consistent. By contrast, errors associated with L2-level fusion are more likely to be amplified in the fusion layer and appear as localized fluctuations or interchannel misalignments. These results suggest that signal-level fusion is more suitable for cross-system and multiplatform applications, providing more reliable inputs for atmospheric retrieval and supporting more robust observations of the middle and upper atmosphere.

V. Conclusion

This study addresses key challenges in dual-channel Rayleigh LiDAR observations for extending the effective dynamic range, including signal saturation, detector nonlinearity, and interchannel inconsistency. A multichannel signal merging and inversion framework is proposed by integrating physics-based constraints with data-driven modeling. The framework combines automated saturation detection, nonlinear physical correction, residual learning, and retrieval-consistent signal fusion before physical inversion. As a result, the continuity and consistency of the merged signals are improved, providing more reliable inputs for subsequent density and temperature retrievals.

Comparative experiments on both isomorphic and heterogeneous PMT systems demonstrate that the proposed method is robust and broadly applicable to density and temperature retrieval. The consistency analysis shows that the L0 strategy, which performs signal-level merging before physical inversion, produces smoother temperature profiles across the transition region and more effectively suppresses error propagation during inversion. By contrast, the L2 strategy, which performs inversion before product-level merging, is more likely to amplify interchannel discrepancies within the overlap region and to introduce local deviations. For the isomorphic configuration, the maximum temperature uncertainty and RMSE within the merging band are 3.7 and 1.62 K for L0, compared with 4.8 and 2.45 K for L2. Under heterogeneous detector configurations, where interchannel

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differences are more pronounced, the advantage of signal-level fusion remains evident. These results indicate that enforcing interchannel consistency at the signal level is important for improving retrieval accuracy and enhancing cross-platform transferability in complex LiDAR systems.

Future work will extend the proposed framework to multiwavelength LiDAR systems with multiple transmission channels, with emphasis on joint multiband fusion and uncertainty propagation. Adaptive fusion and online correction mechanisms will also be further developed to support real-time LiDAR observations.

REFERENCES

[1] Y. Yang et al., "LiDAR network for temperature and wind measurements in the mesosphere and lower thermosphere region," *Space Weather*, vol. 22, no. 10, 2024, Art. no. e2024SW003981.

[2] J. Yan, S. Gong, and Z. Liu, *LiDARs for Environ. Detection*. Beijing, China: Science Press, 2001, pp. 165–167.

[3] Y. Yang et al., "Research progress of LiDAR for upper atmosphere," *Chin. J. Quantum Electron.*, vol. 37, no. 5, pp. 566–579, 2020.

[4] H. Chen, "An overview of the space-based observations for upper atmospheric research," *Adv. Earth Sci.*, vol. 24, no. 3, pp. 229–241, 2009.

[5] X. Cheng et al., "Dual wavelength high altitude detecting LiDAR technology," *Chin. J. Lasers*, vol. 33, no. 5, pp. 601–606, 2006.

[6] C. Xuewu et al., "Na layer and K layer simultaneous observation by LiDAR," *Chin. J. Lasers*, vol. 38, no. 2, 2011, Art. no. 0214001.

[7] D. Yang, T. Zhang, J. Wang, Y. Chunxiao, and H. Peng, "Observations of the prominent sporadic sodium layer over haikou," *Chin. J. Space Sci.*, vol. 38, no. 6, pp. 886–890, 2018.

[8] W. Wang et al., "LiDAR technology and preliminary observation results of wind temperature in the middle and upper atmosphere at Lanzhou Yuzhong station," *Navigat. Control*, vol. 21, p. 250, May 2022.

[9] M. Alpers, R. Eixmann, C. Fricke-Begemann, M. Gerding, and J. Höffner, "Temperature LiDAR measurements from 1 to 105 km altitude using resonance, Rayleigh, and rotational Raman scattering," *Atmos. Chem. Phys.*, vol. 4, no. 3, pp. 793–800, May 2004.

[10] B. Wang et al., "Multichannel interconnection decomposition for hyperspectral LiDAR waveforms detected from over 500 m," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515714.

[11] X. Wang, Z. Ding, L. Dai, N. Li, X. Hao, and Z. Tang, "Rayleigh LiDAR observations of temperature variations in the middle atmopshere over Qujing," in *Proc. 13th Int. Symp. Antennas, Propag. EM Theory (ISAPE)*, Dec. 2021, pp. 1–3.

[12] G. Chen et al., "Hainan coherent scatter phased array radar (HCOPAR): System design and ionospheric irregularity observations," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4757–4765, Aug. 2017.

[13] G. Baumgarten, "Doppler Rayleigh/mie/Raman LiDAR for wind and temperature measurements in the middle atmosphere up to 80 km," *Atmos. Meas. Techn.*, vol. 3, no. 6, pp. 1509–1518, Nov. 2010.

[14] D. P. Donovan and J. A. Whiteway, "Correction for nonlinear photocounting effects in LiDAR systems," *Appl. Opt.*, vol. 32, no. 33, pp. 6742–6753, 1993.

[15] Z. Zhang et al., "Ranging performance model considering the pulse pileup effect for PMT-based photon-counting LiDARs," *Opt. Exp.*, vol. 28, no. 9, pp. 13586–13600, 2020.

[16] T. Luan et al., "Range bias correction for saturated bottom return from bathymetry LiDAR measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5708717, doi: 10.1109/TGRS.2024.3502834.

[17] F. Gao, D. Veberič, S. Stanič, K. Bergant, and D.-X. Hua, "Performance improvement of long-range scanning mie LiDAR for the retrieval of atmospheric extinction," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 122, pp. 72–78, Jun. 2013.

[18] F. Gao et al., "Evaluation and improvement of LiDAR performance based on temporal and spatial variance calculation," *Appl. Sci.*, vol. 9, no. 9, p. 1786, Apr. 2019.

[19] F. Yibin et al., "Signal acquisition system with simultaneous analog and photon counting measurement for LiDAR," *Chin. J. Lasers*, vol. 42, no. 8, 2015, Art. no. 0814001.

[20] L. Wang et al., "Performance of wide dynamic photomultiplier applied in a low blind zone LiDAR," *Remote Sens.*, vol. 15, no. 18, p. 4404, Sep. 2023.

[21] L. Huang, W. Gong, J. Li, F. Mao, and L. Lei, "Signal splicing of dual-receiver mie scattering LiDAR in atmospheric remote sensing," *J. Remote Sens.*, vol. 16, no. 4, pp. 705–719, 2012, doi: 10.11834/jrs.20121217.

[22] Y. Zhang, F. Yi, W. Kong, and Y. Yi, "Slope characterization in combining analog and photon count data from atmospheric LiDAR measurements," *Appl. Opt.*, vol. 53, no. 31, pp. 7312–7320, 2014, doi: 10.1364/ao.53.007312.

[23] C. Feng et al., "Gluing LiDAR signals detected in analog-to-digital and photon counting modes," *EPJ Web Conf.*, vol. 119, Jun. 2016, Art. no. 25021.

[24] C. Feng, Z. Wu, and B. Liu, "Gluing method of detected signal based on LiDAR," *Acta Photon. Sin.*, vol. 47, no. 6, 2018, Art. no. 0601001.

[25] C. Xiang and A. Liang, "Analog and photon signal splicing for CO_2-DIAL based on piecewise nonlinear algorithm," *Atmosphere*, vol. 13, no. 1, p. 109, Jan. 2022.

[26] J. Liu et al., "Analog and photon counting signal gluing for atmospheric LiDAR in cloudy weather," *Opt. Commun.*, vol. 583, Jun. 2025, Art. no. 131729, doi: 10.1016/j.optcom.2025.131729.

[27] T. Wu et al., "An efficient and robust dual-channel signal gluing method for atmospheric LiDAR," *Sensors*, vol. 25, no. 18, p. 5807, Sep. 2025, doi: 10.3390/s25185807.

[28] G. J. Kirchhoff, M. Hayman, W. J. Marais, J. P. Thayer, and R. A. Barton-Grimley, "Development of a photon-counting deadtime noise model that extends dynamic range and resolution in atmospheric LiDAR," *Appl. Opt.*, vol. 64, no. 16, pp. 4568–4581, 2025, doi: 10.1364/ao.543305.

[29] H. Chen et al., "Photon counting method for atmospheric LiDAR echo signals based on waveform decomposition," *Appl. Opt.*, vol. 64, no. 11, pp. 2897–2909, 2025, doi: 10.1364/ao.553318.

[30] Y. Li and X. Xiao, "Deep learning-based fusion of optical, radar, and LiDAR data for advancing land monitoring," *Sensors*, vol. 25, no. 16, p. 4991, Aug. 2025, doi: 10.3390/s25164991.

[31] Y.-Y. Xiang, S. Li, and Y. Ma, "Effect of pile-up of electron flow pulse from photomultiplier tube on ranging by photon counting," *Acta Phys. Sinica*, vol. 71, no. 21, 2022, Art. no. 214206.

[32] D. N. Whiteman, "Examination of the traditional Raman LiDAR technique. I. evaluating the temperature-dependent LiDAR equations," *Appl. Opt.*, vol. 42, no. 15, pp. 2571–2592, 2003.

[33] T. Hakamata et al., *Photomultiplier Tubes: Basics and Applications*, 3rd ed., Hamamatsu, Japan: Hamamatsu Photonics, 2006.

[34] C. A. Cantrell, "Technical note: Review of methods for linear least-squares fitting of data and application to atmospheric chemistry problems," *Atmos. Chem. Phys.*, vol. 8, no. 17, pp. 5477–5487, Sep. 2008.

[35] R. Wang et al., "Simulation research on full waveform acquisition method of high dynamic range LiDAR echo signal," *Opt. Technique*, vol. 50, no. 1, pp. 71–80, 2024.

[36] Y. B. Acharya, S. Sharma, and H. Chandra, "Signal induced noise in PMT detection of LiDAR signals," *Measurement*, vol. 35, no. 3, pp. 269–276, Apr. 2004.

[37] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–20.

[38] S. Chang et al., "Development of a multiscale XGBoost-based model for enhanced detection of potato late blight using Sentinel-2, UAV, and ground data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4415014, doi: 10.1109/TGRS.2024.3466648.

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